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Fluid treatment module

Abstract

A fluid treatment module includes a pipe providing a flow channel in which fluid flows, the pipe having an inlet and an outlet, a light source module including a substrate and at least one light emitting diode disposed on an upper surface of the substrate and emitting light into the pipe to treat the fluid, a reflector disposed inside the pipe to reflect the light emitted from the light source module and having higher reflectivity with respect to the light than the pipe, and a heat dissipation plate contacting a rear surface of the substrate to dissipate heat from the light source module. The inlet, the outlet, or both are provided in plural to allow a flow speed and a flow direction of the fluid flowing into the pipe to be controlled, and the heat dissipation plate has higher thermal conductivity than the substrate.

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Background/Summary

CROSS-REFERENCE OF RELATED APPLICATIONS AND PRIORITY (1) The Present Application is a continuation application of International Application No. PCT/KR2019/014173 filed Oct. 25, 2019 which claims priority to Korean Application No. 10-2018-0129936 filed Oct. 29, 2018, and Korean Application No. 10-2019-0005797 filed Jan. 16, 2019, the disclosures of which are incorporated by reference in their entirety.

TECHNICAL FIELD
(1) Embodiments of the present invention relate to a fluid treatment module.
BACKGROUND
(2) In recent years, as pollution caused by industrialization is increasing, interest in the environment is increasing together with health consciousness. Accordingly, with increasing demand for clean water or clean air, various related products such as water purifiers and air/water purifiers capable of providing clean water and clean air are being developed.
SUMMARY
(3) Embodiments of the present invention provide a device capable of efficiently treating a fluid, such as air or water.
(4) In accordance with one aspect of the present invention, a fluid treatment module includes: a pipe providing a flow channel in which fluid flows, the pipe having an inlet and an outlet; a light source module including a substrate and at least one light emitting diode disposed on an upper surface of the substrate and emitting light into the pipe to treat the fluid; a reflector disposed inside the pipe to reflect the light emitted from the light source module and having higher reflectivity with respect to the light than the pipe; and a heat dissipation plate contacting a rear surface of the substrate to dissipate heat from the light source module. At least one of the inlet and the outlet is provided in plural to allow a flow speed and a flow direction of the fluid flowing into the pipe to be controlled, and the heat dissipation plate has higher thermal conductivity than the substrate.
(5) In one embodiment, the heat dissipation plate may have a larger area than the substrate.
(6) In one embodiment, the heat dissipation plate may be formed of a metal.
(7) In one embodiment, the inlet and the outlet may be provided in different numbers.

(8) In one embodiment, the pipe may include a body extending in a longitudinal direction thereof and first and second ends disposed in the longitudinal direction of the body.
(9) In one embodiment, the heat dissipation plate may have a larger diameter than the body.
(10) In one embodiment, the inlet and the outlet may be disposed at different end sides.
(11) In one embodiment, the inlet and the outlet may be connected to the pipe in the same direction, with a center of the pipe disposed therebetween, when viewed in the longitudinal direction of the pipe.
(12) In one embodiment, the inlet and the outlet may be connected to the pipe in different directions when viewed in the longitudinal direction of the pipe.
(13) In one embodiment, the fluid treatment module may further include a reflector disposed inside the pipe and reflecting light emitted from the light source module.
(14) In one embodiment, the reflector may have higher reflectivity with respect to the light than the pipe.
(15) In one embodiment, the reflector may include a first reflector disposed on an inner wall of the pipe and a second reflector disposed on the substrate of the light source module. In one embodiment, the second reflector may have an opening exposing the light emitting diode and an inner surface of the opening may be an inclined surface.
(16) In one embodiment, the reflector may be formed of a porous material and may have a reflectivity of 80% or more.
(17) In one embodiment, the inlet may be provided as a pair and the outlet may be provided singularly.
(18) In one embodiment, the inlet and the outlet may have the same diameter or different diameters.
(19) In one embodiment, the fluid treatment module may be employed by a water supply device, which includes a reservoir receiving water and a water treatment device connected to the reservoir and treating the water.
(20) Embodiments of the present invention provide a fluid treatment module having high treatment efficiency and high reliability.

Description

DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a perspective view of a fluid treatment module according to an embodiment of the present invention.
(2) FIG. **2** is an exploded perspective view of the fluid treatment module according to the embodiment of the present invention.
(3) FIG. **3** is a perspective longitudinal sectional view of the fluid treatment module shown in FIG. **1**.
(4) FIG. **4**A is a sectional view of some components of the fluid treatment module according to the embodiment of the present invention, particularly illustrating a pipe, thereof.
(5) FIG. **4**B is a sectional view of some components of the fluid treatment module according to the embodiment of the present invention, particularly illustrating an inlet thereof.
(6) FIG. **4**C is a sectional view of some components of the fluid treatment module according to the embodiment of the present invention, particularly illustrating an outlet thereof.
(7) FIG. **5**A is a side sectional view of the fluid treatment module according to the embodiment of the present invention, particularly illustrating the pipe thereof.
(8) FIG. **5**B is a side sectional view of the fluid treatment module according to the embodiment of the present invention, particularly illustrating the inlet thereof.
(9) FIG. **5**C is a side sectional view of the fluid treatment module according to the embodiment of the present invention, particularly illustrating the outlet thereof.

(10) FIG. **6** is an exploded perspective view of a fluid treatment module according to an embodiment of the present invention.
(11) FIG. **7** is a perspective longitudinal sectional view of the fluid treatment module according to the embodiment of the present invention.
(12) FIG. **8**A is a graph depicting a fluid treatment effect depending upon the presence of a reflector with respect to Examples 1 and 2.
(13) FIG. **8**B is a graph depicting a fluid treatment effect depending upon the presence of a reflector with respect to Examples 3 and 4.
(14) FIG. **8**C is a side sectional view of a fluid treatment module used in experiments of FIG. **8**A and FIG. **8**B.
(15) FIG. **9**A is a side view of a fluid treatment module provided with an inlet and an outlet through a first modification.
(16) FIG. **9**B is a side view of a fluid treatment module provided with an inlet and an outlet through a second modification.
(17) FIG. **9**C is a side view of a fluid treatment module provided with an inlet and an outlet through a third modification.
(18) FIG. **9**D is a side view of a fluid treatment module provided with an inlet and an outlet through a fourth modification.
(19) FIG. **9**E is a side view of a fluid treatment module provided with an inlet and an outlet through a fifth modification.
(20) FIG. **9**F is a side view of a fluid treatment module provided with an inlet and an outlet through a sixth modification.
(21) FIG. **10**A is a graph depicting fluid treatment effects of the fluid treatment modules shown in FIG. **9**A.
(22) FIG. **10**B is a graph depicting fluid treatment effects of the fluid treatment modules shown in FIG. **9**B.
(23) FIG. **10**C is a graph depicting fluid treatment effects of the fluid treatment modules shown in FIG. **9**C.
(24) FIG. **10**D is a graph depicting fluid treatment effects of the fluid treatment modules shown in FIG. **9**D.
(25) FIG. **10**E is a graph depicting fluid treatment effects of the fluid treatment modules shown in FIG. **9**E.
(26) FIG. **10**F is a graph depicting fluid treatment effects of the fluid treatment modules shown in FIG. **9**F.
(27) FIG. **11** is a graph depicting sterilization efficiency depending upon a flow rate of a fluid flowing in the pipe of the fluid treatment module according to the embodiment of the present invention.
(28) FIG. **12**A is a graph depicting sterilization efficiency depending upon the flow rate of the fluid flowing in the pipe of the fluid treatment module according to the embodiment of the present invention.
(29) FIG. **12**B is a graph depicting sterilization efficiency depending upon a different flow rate of the fluid flowing in the pipe of the fluid treatment module according to the embodiment of the present invention.
(30) FIG. **13** is a schematic view of a water purifier adopting the fluid treatment module according to the embodiment of the present invention.

DETAILED DESCRIPTION OF EMBODIMENTS

(31) Hereinafter, embodiments of the present disclosure will be described with reference to the accompanying drawings. It should be understood that the embodiments are provided for complete disclosure and a thorough understanding of the present disclosure by those skilled in the art. Therefore, the present disclosure is not limited to the following embodiments and may be

embodied in different ways. In addition, the drawings may be exaggerated in width, length, and thickness of components for descriptive convenience and clarity only. Like components will be denoted by like reference numerals throughout the specification.
(32) Hereinafter, exemplary embodiments of the present invention will be described in detail with reference to the accompanying drawings so as to fully convey the spirit of the present invention to those skilled in the art.
(33) FIG. **1** is a perspective view of a fluid treatment module according to an embodiment of the present invention and FIG. **2** is an exploded perspective view of the fluid treatment module according to the embodiment of the present invention. FIG. **3** is a perspective longitudinal sectional view of the fluid treatment module shown in FIG. **1**.
(34) One embodiment of the present invention relates to a fluid treatment module. In the embodiment, the term “fluid” refers to an object to be treated using the fluid treatment module and may include water (especially running water) or air. In one embodiment, treatment of the fluid includes, for example, sterilizing, purifying, and deodorizing the fluid through the fluid treatment module. However, treatment of the fluid is not limited thereto and may include other possible treatments carried out using the fluid treatment module described below.
(35) Referring to FIG. **1** to FIG. **3**, a fluid treatment module **100** according to one embodiment includes a pipe **110**, in which fluid flows, and a light source module **130** emitting light towards the fluid in the pipe **110**. The pipe **110** extends in one direction and has a shape open at opposite ends thereof. The pipe **110** may include a body **110***c*, which extends in a longitudinal direction thereof, and a first end **110***a* and a second end **110***b* disposed in the longitudinal direction of the body **110***c*. The body **110***c* has a hollow pipe shape having a predetermined diameter. The first end **110***a* and the second end **110***b* are connected to the opposite ends of the body **110***c* and may have the same diameter as or a larger diameter than the body **110***c*. The pipe **110** provides an interior space, that is, a flow channel, in which the fluid flows while being treated. Hereinafter, a direction in which the pipe **110** extends will be referred to as an extension direction of the pipe **110** or the longitudinal direction of the pipe **110**.
(36) In the embodiment, the pipe **110** may have a substantially cylindrical shape. In this case, a cross-section crossing a longitudinal direction of the cylindrical shape has a circular shape. However, it should be understood that the pipe **110** is not limited thereto and may have various cross-sectional shapes, for example, an elliptical shape, a rectangular shape, a semicircular shape, and the like.
(37) The pipe **110** may be formed of a material having high reflectivity and/or a metal having high thermal conductivity. For example, the pipe **110** may be formed of a material having high reflectivity, such as stainless steel, aluminum, magnesium oxide, and the like, or may be formed of a material having high thermal conductivity, such as stainless steel, aluminum, silver, gold, copper, and alloys thereof. Metals having high thermal conductivity can effectively discharge heat from the pipe **110**.
(38) However, the pipe **110** is not limited thereto and may be formed of a material that transmits at least part of light emitted from the light source module **130** such that the light can reach the fluid inside the pipe **110**, when the light source module **130** is disposed outside the pipe **110**. The material capable of transmitting light may include various types of polymer resins, quartz, glass, and the like. For example, the entirety of the pipe **110** or a portion of the pipe **110** adjacent to the light source module **130** may be formed of a transparent material to allow the light emitted from the light source module **130** to reach the fluid.
(39) The pipe **110** is provided with an inlet **113** through which the fluid enters the pipe **110** and an outlet **115** through which the fluid is discharged therefrom after treatment. The inlet **113** and/or the outlet **115** may be provided to at least one region of the pipe **110** selected from among the body **110***c*, the first end **110***a*, and the second end **110***b*.
(40) As shown in FIG. **1** and FIG. **2**, the inlet **113** is connected to the pipe **110**. A connecting

direction of the inlet **113** may be different from the extension direction of the pipe **110**. In one embodiment, the connecting direction of the inlet **113** may be inclined or perpendicular to the extension direction of the pipe **110** such that the fluid enters the pipe **110** in a direction inclined or perpendicular thereto and flows in the extension direction of the pipe **110**. The fluid flowing into the pipe **110** through inlet **113** is a fluid to be treated in the pipe **110**, for example, an object that requires sterilization, purification, or deodorization treatment.

(41) The outlet **115** may be disposed at a location separated from the inlet **113** and may be connected to the pipe **110**. In one embodiment, a connecting direction of the outlet **115** may be inclined or perpendicular to the extension direction of the pipe **110** such that the fluid can be discharged from the pipe **110** in a direction inclined or perpendicular thereto after flowing in the extension direction of the pipe **110**. The fluid discharged from the pipe **110** through the outlet **115** is a fluid treated in the pipe **110**, for example, an object subjected to sterilization, purification, or deodorization treatment.

(42) In one embodiment, when viewed in a direction perpendicular to the longitudinal direction of the pipe **110**, the inlet **113** may be provided to at least one end side selected from among the opposite ends of the pipe **110** and the outlet **115** may also be provided to at least one end side selected from among the opposite ends of the pipe **110**. In other words, in the embodiment, assuming the opposite ends of the pipe **110** in the longitudinal direction are referred to as the first end **110***a* and the second end **110***b*, respectively, each of the inlet **113** and the outlet **115** may be provided to at least one end side selected from among the first end **110***a* and the second end **110***b* or may be provided to both the first end **110***a* side and the second end **110***b* side. For example, as shown in the drawings, the inlet **113** may be provided to the first end **110***a* side and the outlet **115** may be provided to the second end **110***b* side. Alternatively, the inlet **113** may be provided to the second end **110***b* side and the outlet **115** may be provided to the first end **110***a* side. However, it should be understood that the locations of the inlet **113** and the outlet **115** are not limited thereto. For example, the inlet **113** and/or the outlet **115** may be disposed at a center of the pipe **110** instead of the opposite ends thereof.

(43) In the embodiment, the inlet **113** and the outlet **115** may be disposed in various ways when viewed in the longitudinal direction of the pipe **110**. FIG. **4**A to FIG. **4**C are sectional views of some components of the fluid treatment module according to the embodiment of the present invention, particularly illustrating the pipe **110**, the inlet **113**, and the outlet **115**, in which each of the inlet **113** and the outlet **115** is provided singularly. In FIG. **4**A to FIG. **4**C, reference numeral **121** indicates a first reflector.

(44) Referring to FIG. **4**A, the inlet **113** and the outlet **115** may be connected to the pipe **110** to face each other with the center of the pipe **110** disposed therebetween, when viewed in a cross-section perpendicular to the extension direction of the pipe **110**. That is, the inlet **113** and the outlet **115** may be disposed at symmetric locations with the center of the pipe **110** disposed therebetween. However, it should be understood that the inlet **113** and the outlet **115** are not required to be completely symmetric about the center of the pipe **110**. Referring to FIG. **4**B, the inlet **113** and the outlet **115** may be connected to the pipe **110** at the same side with reference to the center of the pipe **110**. In this case, as shown in the drawings, the inlet **113** and the outlet **115** may be separated from the pipe **110**. That is, the inlet **113** and the outlet **115** may be collinearly or non-collinearly disposed in the extension direction of the pipe **110**. Alternatively, the inlet **113** and the outlet **115** may be disposed to overlap each other. Referring to FIG. **4**C, the inlet **113** and the outlet **115** may be connected to the pipe **110** to be perpendicular to each other.

(45) The locations of the inlet **113** and the outlet **115** may be changed depending upon the kind of apparatus adopting the fluid treatment module. In particular, the locations of the inlet and the outlet may be set in various ways depending upon a fluid treatment amount or a degree of fluid sterilization required by the apparatus.

(46) Although each of the inlet **113** and the outlet **115** is provided singularly in FIGS. **1-3**, it should

be understood that this structure is provided for illustration of a positional relationship between the inlet **113** and the outlet **115**. Accordingly, when one of the inlet **113** and the outlet **115** is provided in plural, the inlets **113** or the outlets **115** may be individually disposed at different locations. For example, for the fluid treatment module including two inlets **113**, that is, a first inlet **113***a* and a second inlet **113***b*, the first inlet **113***a* is disposed at a side opposite the outlet **115** and the second inlet **113***b* may be disposed at the same side as the outlet **115** when viewed in the extension direction of the pipe **110**.
(47) Referring again to FIG. **1** to FIG. **3**, in the embodiment, the inlet **113** and the outlet **115** may have a circular or elliptical cross-section, without being limited thereto. Alternatively, the inlet **113** and the outlet **115** may have various shapes, for example, a polygonal shape. Here, the cross-section of each of the inlet **113** and the outlet **115** may be taken in the extension direction of the inlet **113** or in a direction crossing a direction in which the flow channel is formed.
(48) Although not shown in the drawings, the inlet **113** and/or the outlet **115** may be provided with a separate pipe. The separate pipe may be connected to the inlet **113** and the outlet **115** through a nozzle. The nozzle may be coupled to the inlet **113** and/or the outlet **115** in various ways, for example, by screw coupling.
(49) In one embodiment, the flow amount or speed of the fluid supplied through the inlet **113** and the flow amount or speed of the fluid discharged through the outlet **115** may be controlled depending upon the number and diameter of the inlets **113** and the outlets **115**. To this end, in one embodiment, the inlet **113** and the outlet **115** may have the same diameter or different diameters. According to the embodiment, although the inlet **113** having the same diameter as the outlet **115** can provide good fluid treatment efficiency, the inlet **113** having a larger diameter than the outlet **115** can provide better fluid treatment efficiency.
(50) Further, in one embodiment, at least one of the inlet **113** and the outlet **115** may be provided in plural. For example, the inlet **113** may be provided in plural and the outlet **115** may be provided singularly. Alternatively, the inlet **113** may be provided singularly and the outlet **115** may be provided in plural or both the inlet **113** and the outlet **115** may be provided in plural. For the fluid treatment device including multiple inlets **113** and multiple outlets **115**, the inlets **113** and the outlets **115** may be provided in one-to-one correspondence. In particular, in one embodiment, the fluid treatment device including two inlets **113** and one outlet **115** can have improved fluid treatment efficiency.
(51) FIG. **5**A to FIG. **5**C are side sectional views of the fluid treatment module according to the embodiment of the present invention, particularly illustrating the pipe, the inlet and the outlet thereof.
(52) In one embodiment, as shown in FIG. **5**A to FIG. **5**C, each of the inlet and the outlet may be provided in various numbers. For example, the fluid treatment module may include multiple inlets (that is, a first inlet **113***a* and a second inlet **113***b*) and a single outlet **115**, as shown in FIG. **5**A; a single inlet **113** and multiple outlets (that is, a first outlet **115***a* and a second outlet **115***b*), as shown in FIG. **5**B; or multiple inlets (that is, a first inlet **113***a* and a second inlet **113***b*) and multiple outlets (that is, a first outlet **115***a* and a second outlet **115***b*), as shown in FIG. **5**C.
(53) Although two inlets and two outlets are illustrated by way of example, it should be understood that the present invention is not limited thereto, and a greater number of inlets or outlets may be provided.
(54) Referring again to FIG. **1** to FIG. **3**, for the fluid treatment module including two inlets **113** according to the embodiment, when viewed in the direction perpendicular to the extension direction of the pipe **110**, the first and second inlets **113***a*, **113***b* may be provided to the second end **110***b* side and the outlet **115** may be provided to the first end **110***a* side, and when viewed in the extension direction of the pipe **110**, the first inlet **113***a* may be disposed at an opposite side to the outlet **115**, with the center of the pipe **110** disposed therebetween, and the second inlet **113***b* may be disposed at the same side as the outlet **115** with reference to the center of the pipe **110**.

(55) The fluid treatment effect depending upon the number and diameter of the inlets **113** and the outlets **115** will be described below.
(56) The light source module **130** supplies light suitable for treatment of the fluid. The light source module **130** may be disposed at various locations adjacent to the fluid to emit light for treatment (for example, sterilization, purification, or deodorization) of the fluid.
(57) The light source module **130** emits light and may be disposed at one side selected from among the first end **110***a* and the second end **110***b* of the pipe **110**. In this embodiment, the light source module **130** is provided to both the first end **110***a* and the second end **110***b* by way of example. However, it should be understood that the location of the light source module **130** is not limited thereto and may be provided to one of the opposite ends of the pipe **110** in the longitudinal direction thereof. It should be understood that the location of the light source module **130** in this embodiment is provided for illustration only and is not to be construed in any way as limiting the present invention and that the light source module **130** may be disposed at any location so long as the light source module **130** can irradiate the interior of the pipe **10**. Unlike in FIGS. **2-3**, the light source module **130** may be disposed outside the pipe **110**, without being limited thereto.
(58) The light source module **130** may include a substrate **131** and a light emitting diode **133** mounted on the substrate **131**. The substrate **131** may be provided in various forms, for example, in the form of a disk having a diameter corresponding to the pipe **10**. Multiple light emitting diodes **133** may be arranged in a predetermined direction on the substrate **131**. The substrate **31** may be provided with an opening through which a wire is connected to the light emitting diode **133** to supply power thereto.
(59) For the light source module **130** including multiple light emitting diodes **133**, the light emitting diodes **133** may emit light in the same wavelength band or in different wavelength bands. For example, in one embodiment, all of the light emitting diodes **133** may emit UV light having the same or similar wavelength. In another embodiment, some light emitting diodes **133** may emit light in some UV wavelength bands and the other light emitting diodes **133** may emit light in other UV wavelength bands.
(60) When the light emitting diodes **133** emit light in different wavelength bands, the light emitting diodes **133** may be arranged in various sequences. For example, light emitting diodes **133** emitting light in a first wavelength band and light emitting diodes **133** emitting light in a second wavelength band different from the first wavelength band may be alternately arranged.
(61) The light source module **130** may emit light in various wavelength bands. The light source module **130** may emit light in the visible spectrum, light in the UV spectrum, or light in other wavelength bands. In one embodiment, light emitted from the light source module **130** may have various wavelengths depending upon the type of fluid to be treated or the type of object to be treated (for example, germs, bacteria, and the like). In particular, when the fluid is to be sterilized, the light source module **130** may emit light having a germicidal wavelength. For example, the light source module **130** may emit light in the UV spectrum. In one embodiment, the light source module **130** may emit light in the wavelength band of about 100 nm to about 420 nm, which is germicidal to microorganisms. In one embodiment, the light source module **130** may emit light in the wavelength band of about 100 nm to about 280 nm. In another embodiment, the light source module **130** may emit light in the wavelength band of about 180 nm to about 280 nm. In a further embodiment, the light source module **130** may emit light in the wavelength band of about 250 nm to about 260 nm. UV light in the above wavelength bands has high germicidal efficacy. For example, irradiation with UV light at an intensity of 100 μW/cm.sup.2 can kill about 99% of bacteria, such as *Escherichia coli, Diphtheria bacillus*, and *Dysentery bacillus*. In addition, UV light in the above wavelength bands can kill bacteria that cause food poisoning, such as pathogenic *Escherichia coli, Staphylococcus aureus, Salmonella Weltevreden, S. Typhimurium, Enterococcus faecalis, Bacillus cereus, Pseudomonas aeruginosa, Vibrio parahaemolyticus, Listeria monocytogenes, Yersinia enterocolitica, Clostridium perfringens, Clostridium botulinum,*

Campylobacter jejuni, or *Enterobacter sakazakii*.

(62) In one embodiment, the light source module **130** may emit light having various wavelengths and at least part of the light source module **130** may include a material that causes a catalytic reaction in response to light emitted from the light source module **130**. For example, a photocatalytic layer formed of a photocatalytic material may be disposed on the entirety or a portion of an inner surface and/or an outer surface of the pipe **10** according to the present invention. The photocatalytic layer may be disposed in any region so long as the light emitted from the light source module **130** can reach the region.

(63) A photocatalyst refers to a material causing a catalytic reaction with light emitted from a light source. The photocatalyst can react with light in various wavelength bands depending on substances constituting the photocatalyst. In one embodiment, a material causing photocatalytic reaction with light in the UV wavelength band among various wavelength bands may be used. However, the photocatalyst is not limited thereto and other photocatalysts having the same or similar mechanism may be used depending on light emitted from the light source.

(64) The photocatalyst is activated by UV light to cause chemical reaction, thereby decomposing various pollutants and bacteria in the fluid, which contacts the photocatalyst, through redox reaction.

(65) Although not shown in FIGS. **1-3**, the fluid treatment module **100** according to the embodiment may further include a drive circuit connected to the light source module **130** and an interconnect portion connecting the drive circuit to the light source module **130**. The drive circuit may supply electric power to at least one light source module **130**. For example, the drive circuit may be provided to the fluid treatment module **100** having two light source modules **130** to independently supply electric power to each of the light source modules **130**. Accordingly, the light source modules **130** may be selectively driven such that all of the two light source modules **130** can be turned on or off, or one light source module **130** can be turned on, with the other light source module **130** turned off.

(66) A transmissive window **137** may be further disposed between the light source module **130** and a fluid treatment space inside the pipe **110** to transmit light emitted from the light source module **130** to the interior of the pipe **110**.

(67) The transmissive window **137** serves to protect the substrate **131** and the light source and may be formed of a transparent, electrically insulating material. However, it should be understood that the present invention is not limited thereto and the transmissive window **137** may be formed of various other materials. For example, the transmissive window **37** may be formed of quartz or an organic polymer. Here, since the wavelength of light absorbed by/transmitted through the organic polymer depends on the type of monomers for the organic polymer, a method of forming the organic polymer, and conditions in which the organic polymer material is formed, the organic polymer may be selected in consideration of wavelengths of light emitted from the light sources. For example, organic polymers, such as poly (poly methyl methacrylate; PMMA), polyvinyl alcohol (PVA), polypropylene (PP), and low-density polyethylene (PE), absorb little or no UV light, whereas polymer resins such as polyester can absorb UV light.

(68) In this embodiment, the substrate **131** and the transmissive window **137** may correspond to the pipe **110** in terms of shape and size.

(69) A heat dissipation plate is disposed on an outer surface of the light source module **130** to dissipate heat generated from the light source module **130**. In one embodiment, in a structure where the light source modules **130** are provided to the opposite ends of the pipe **110**, a first heat dissipation plate **140***a* and a second heat dissipation plate **140***b* are disposed outside the two light source modules **130**, respectively.

(70) The first and second heat dissipation plates **140***a*, **140***b* are formed of a material having higher thermal conductivity than the substrate **131** of the light source module **130**, and serve to discharge heat, which is generated from the light source module **130**, particularly from the light emitting

diode **133**, and is transferred through the substrate **131**. To this end, each of the first and second heat dissipation plates **140***a*, **140***b* directly contacts the rear surface of the substrate **131** of the light source module **130**.
(71) As the material having higher thermal conductivity than the substrate **131** of the light source module **130**, various materials may be used. For the substrate **131** formed of a metal, each of the first and second heat dissipation plates **140***a*, **140***b* may be formed of a material having higher thermal conductivity than the metal constituting the substrate **131**. For example, when the substrate **131** is formed of stainless steel, the first and second heat dissipation plates **140***a*, **140***b* may be formed of a metal, for example, aluminum, which has higher thermal conductivity than stainless steel. The material for the first and second heat dissipation plates **140***a*, **140***b* may be selected from among various materials so long as the first and second heat dissipation plates have higher thermal conductivity than the substrate **131**, instead of being limited to a particular material.
(72) In one embodiment, the first and second heat dissipation plates **140***a*, **140***b* may have a larger area than the substrate **131** to allow efficient dissipation of heat from the substrate **131**. The substrate **131** is disposed inside the pipe **110** and may have a diameter corresponding to the pipe **110**, and the first and second heat dissipation plates **140***a*, **140***b* may have a larger diameter than the substrate **131**. In addition, each of the first and second heat dissipation plates **140***a*, **140***b* may have a larger diameter than the body **110***c* of the pipe **110**. Since the first and second heat dissipation plates **140***a*, **140***b* may disposed at the opposite end sides of the pipe **110**, that is, at the first and second ends **110***a*, **110***b*, the first and second heat dissipation plates **140***a*, **140***b* can be formed to protrude from the opposite ends of the pipe **110** and can be easily formed to have larger diameters.
(73) In the embodiment, the first and second heat dissipation plates **140***a*, **140***b* enable easier dissipation of heat from a light source. As a result, the light source is prevented from being deteriorated due to heat from the light source, thereby providing a stable sterilization effect of the fluid treatment module **100** while improving reliability thereof.
(74) In one embodiment, the fluid treatment module **100** may be provided with a reflector **120** in addition to the substrate **131** and the light emitting diodes **133** to improve fluid treatment efficiency by reflecting light emitted from the light emitting diodes **133** to travel inside the pipe **110**.
(75) The reflector **120** has higher reflectivity than the pipe **110**. For example, when the pipe **110** is formed of a material, such as stainless steel, the reflector **120** is formed of a material having higher reflectivity than the stainless steel. In one embodiment, the reflector **120** formed of a material having higher reflectivity than the pipe **100** may be disposed inside the pipe **110** to increase reflectivity with respect to light traveling inside the pipe **110**. Increase in reflectivity of light results in improvement in fluid treatment efficiency through efficient reflection of light inside the pipe **110**.
(76) The reflector **120** may be formed of a material that reflects light emitted from the light source module **130**. The reflector **120** may be formed of a material reflecting 80% or more of light emitted from the light source module **130**, and in one embodiment, 90% or more of the light in another embodiment, or 99% or more of the light in a further another embodiment.
(77) The reflector **120** may be formed of a material having high reflectivity to maximize efficiency in extraction of light emitted from the light emitting diode chip **133**. For example, the reflector **120** may be formed of a material having roughness to improve reflection and scattering of light on the surface thereof. Alternatively, the reflector **120** may be formed of a material having a porous surface. Although there are various porous materials having surface roughness, the reflector according to one embodiment may be formed of a polymer resin, for example, PTFE. However, it should be understood that the reflector **120** is not limited thereto and may be formed of other materials so long as the reflector **120** can secure sufficient reflectivity. For example, the reflector **120** may be formed of aluminum and/or an aluminum alloy. Alternatively, the reflector **120** may be formed of a material having high reflectivity, for example, various metals, such as silver, gold, tin, copper, chromium, nickel, molybdenum, titanium, and the like, and/or alloys thereof.

(78) The reflector **120** may include a first reflector **121** disposed inside the pipe **110** and a second reflector **123** disposed near the light source module **130**. The first reflector **121** and the second reflector **123** may be formed of the same material or different materials. For example, both the first reflector **121** and the second reflector **123** may be formed of polytetrafluoroethylene (PTFE). Alternatively, the first reflector **121** may be formed of PTFE and the second reflector **123** may be formed of an aluminum alloy.
(79) The first reflector **121** covers the inner surface of the pipe **110**. For the pipe **110** having a cylindrical shape, the first reflector **121** may be formed to cover the entirety of a cylindrical inner wall of the pipe **110**. Here, the first reflector **121** may be present or absent on portions of the inner wall of the pipe **110** corresponding to the inlet **113** and the outlet **115**. In the embodiment, the inner wall of the pipe **110** may be substantially prevented from being exposed by the first reflector **121**, which is disposed on the inner wall of the pipe **110** so as to cover a light reaching region as much as possible.
(80) The second reflector **123** is disposed on the substrate **131** along the periphery of a mounting region of the light emitting diodes **133**. The second reflector **123** serves to reflect light emitted from the light emitting diodes **133** to travel toward each region in the pipe **110**. To this end, the second reflector **123** has an opening exposing the mounting region of the light emitting diodes **133** and is disposed on the substrate **131** in a ring shape penetrated through upper and lower portions thereof.
(81) According to one embodiment, the second reflector **123** may have an inner surface facing the opening, an outer surface facing the outside, and a bottom surface adjoining an upper surface of the substrate **131**. The inner surface is at least partially inclined with respect to the upper surface of the substrate **131**. With this structure, the opening of the second reflector **123** has a width gradually increasing from the upper surface of the substrate **131** in an upward direction. In other words, the second reflector **123** has an inner diameter gradually increasing from the upper surface of the substrate **131** in the upward direction. In a cross-section of the second reflector **123**, an inclined side may be linear or curved and may have an inclination set at various angles in consideration of the number of light emitting diodes **133**, a beam angle of the light emitting diode **133**, and the intensity of light emitted from the light emitting diode **133**, and the like.
(82) The fluid treatment module **100** according to this embodiment may be provided with one or more sealing members **151***a*, **151***b* to tightly fasten the pipe **10** to the first and second heat dissipation plates **140***a*, **140***b* while preventing the fluid from leaking out of the fluid treatment module **100**.
(83) In one embodiment, the sealing members **151***a*, **151***b* may be disposed near a fluid treatment space, for example, between a first cap and the first end **110***a* of the pipe **110** and between a second cap and the second end **110***b* of the pipe **110**, respectively. The first cap and the second cap will be further described below. Each of the sealing members **151***a*, **151***b* may include first and second sealing members **150***a*, **150***b* disposed between the substrate **131** and the transmissive window **137** and between the transmissive window **137** and a stepped portion inside the pipe **110**, respectively. The first and second sealing members **150***a*, **150***b* serve to tightly fasten the pipe **110** to the first and second caps while preventing the fluid in a first space from leaking out through a gap between the pipe **110** and the first and second caps. For example, the first and the second sealing members **150***a*, **150***b* separate the fluid treatment space from the light source module **130** to prevent the light source module **130** from being damaged by the fluid. Each of the sealing members **151***a*, **151***b* may be provided singularly or in plural.
(84) Each of the sealing members **151***a*, **151***b* has a closed cross-section so as to tightly fasten an inner region and an outer region of the pipe **110** to each other and to isolate and seal these regions from each other when the first and second caps are fastened to the pipe **110**. For example, each of the first and second sealing members **150***a*, **150***b* may be provided in the form of an O-ring.
(85) The sealing members **151***a*, **151***b* may be formed of a soft elastic material. When the sealing

members **151***a*, **151***b* are formed of such an elastic material, the sealing members **151***a*, **151***b* can be compressed against the pipe **110** upon fastening the pipe **110** to the first and second heat dissipation plates **140***a*, **140***b*, thereby maintaining a tight fastening structure.
(86) Although the elastic material forming the sealing members **151***a*, **151***b* may include a silicone resin, it will be understood that the present invention is not limited thereto and the sealing members **151***a*, **151***b* may be formed of any other suitable material. For example, natural or synthetic rubber or other elastic organic polymers may be used as the elastic material.
(87) In one embodiment, the first and second heat dissipation plates **140***a*, **140***b* may be used as first and second caps encapsulating the first and second ends **110***a*, **110***b* of the pipe **110**, respectively. Accordingly, in this embodiment, each of the first and second heat dissipation plates **140***a*, **140***b* may have a fastening portion coupled to the pipe **110**.
(88) The fastening portion may be provided in various forms. For example, as the fastening portion, each of the first and second heat dissipation plates **140***a*, **140***b* may have an insertion portion, which has a diameter corresponding to the inner diameter of the pipe **110** and is inserted into an end of the pipe **110** to encapsulate the pipe **110**.
(89) In one embodiment, the first heat dissipation plate **140***a* is disposed at the first end **110***a* of the pipe **110** to be fastened to the pipe **110**. The first heat dissipation plate **140***a* is formed with stepped portions having different outer diameters to be inserted into and fastened to the pipe **110**. For example, a portion of the first heat dissipation plate **140***a* facing the first end **110***a* of the pipe **110** has an outer diameter corresponding to the inner diameter of the pipe **110**.
(90) In one embodiment, the pipe **110** and the first heat dissipation plate **140***a* may be provided with fastening members for coupling the pipe **110** to the first heat dissipation plate **140***a*. For example, the pipe **110** and the first heat dissipation plate **140***a* may be provided with fastening holes **143***a* such that the pipe **110** can be coupled to the first heat dissipation plate **140***a* by coupling fastening bolts **143***b* to the fastening holes **143***a* of the pipe **110** and the first heat dissipation plate **140***a*.
(91) The second heat dissipation plate **140***b* is provided to the second end **110***b* of the pipe **110** to be fastened to the pipe **110**. The second heat dissipation plate **140***b* may be also formed with stepped portions having different outer diameters and may be inserted into and fastened to the pipe **110** in the same manner as the first heat dissipation plate **140***a*.
(92) Likewise, the pipe **110** and the second heat dissipation plate **140***b* may be provided with fastening members for coupling the pipe **110** to the second heat dissipation plate **140***b*. For example, the pipe **110** and the second heat dissipation plate **140***b* may be provided with fastening holes **143***a* such that the pipe **110** can be coupled to the second heat dissipation plate **140***b* by coupling fastening bolts **143***b* to the fastening holes **143***a* of the pipe **110** and the second heat dissipation plate **140***b*.
(93) In one embodiment, the first and second heat dissipation plates **140***a*, **140***b* are used not only as heat dissipation members for dissipation of heat generated from the light source module **130**, but also as tools for encapsulating the pipe **110**, that is, as the first and second caps. Alternatively, encapsulating members for encapsulating the pipe **110** may be further provided as separate components from the first and second heat dissipation plates **140***a*, **140***b*. In this case, the separate encapsulating members may also be formed of a material having high thermal conductivity to allow efficient dissipation of heat from the first and second heat dissipation plates **140***a*, **140***b*.
(94) In the fluid treatment module with the above structure, as the fluid flows in the extension direction of the pipe, the fluid is subjected to treatment, such as sterilization and the like, through exposure to light emitted from the light source module. The fluid treatment module according to the embodiments achieves remarkable improvement in fluid treatment efficiency through control of the number and diameter of the inlets and the outlets while improving treatment efficiency with light through the reflector. In addition, the fluid treatment module adopts the heat dissipation plate, thereby improving reliability of the light source module.

(95) The fluid treatment module according to the embodiments of the present invention may be modified in various shapes without departing from the concept of the present invention.
(96) FIG. **6** is an exploded perspective view of a fluid treatment module according to an embodiment of the present invention and FIG. **7** is a perspective longitudinal sectional view of the fluid treatment module according to the embodiment of the present invention. The following description will focus on different features from the above embodiments to avoid repetition of description.
(97) Referring to FIG. **6** and FIG. **7**, a fluid treatment module **100** according to the embodiment includes a pipe **110**, in which fluid flows, and a light source module **130** emitting light towards the fluid in the pipe **110**. The pipe **110** extends in one direction and has a shape closed at one end thereof and open at the other end thereof. For the pipe **110** including a body **110***c* extending in a longitudinal direction thereof and a first end **110***a* and a second end **110***b* disposed in the longitudinal direction of the body **110***c*, the first end **110***a* has a closed shape having no inlet and the second end **110***b* has an open shape.
(98) The pipe **110** is provided with an inlet **113** through which the fluid enters the pipe **110** and an outlet **115** through which the fluid is discharged therefrom after treatment. In this embodiment, the inlet **113** may be provided to an open-end side, that is, to the second end **110***b* side, and the outlet **115** may be provided to a closed end side, that is, to the first end **110***a* side.
(99) In this embodiment, the inlet **113** may be provided in plural, for example, as a first inlet **113***a* and a second inlet **113***b*. In this embodiment, both the first inlet **113***a* and the second inlet **113***b* may be provided to the second end **110***b* side.
(100) In this embodiment, the flow amount or speed of the fluid supplied through the inlet **113** and the flow amount or speed of the fluid discharged through the outlet **115** may be controlled depending upon the number and diameter of the inlets **113** and the outlets **115**, as in the above embodiment.
(101) The light source module **130** supplies light suitable for treatment of the fluid. In this embodiment, the light source module **130** may be disposed at the second end **110***b* side among the first end **110***a* and the second end **110***b* of the pipe **110**. The light source module **130** is not disposed at the first end **110***a* side, which does not include an opening. The light source module **130** may include a substrate **131** and a light emitting diode **133** mounted on the substrate **131**.
(102) A transmissive window **137** may be further disposed between the light source module **130** and a fluid treatment space inside the pipe **110** to transmit light emitted from the light source module **130** to the interior of the pipe **110**.
(103) A heat dissipation plate **140** is disposed at an open-end side of the light source module **130** to dissipate heat generated from the light source module **130**. In a structure where the light source module **130** is provided to the second end **110***b* of the pipe **110**, the heat dissipation plate **140** is provided at the second end **110***b* side.
(104) The heat dissipation plate **140** is formed of a material having higher thermal conductivity than the substrate **131** of the light source module **130**, and serves to discharge heat, which is generated from the light source module **130**, particularly from the light emitting diode **133**, and is transferred through the substrate **131**. To this end, the heat dissipation plate **140** directly contacts the rear surface of the substrate **131** of the light source module **130**.
(105) In one embodiment, the heat dissipation plate **140** may have a larger area than the substrate **131** to allow efficient dissipation of heat from the substrate **131**. The substrate **131** is disposed inside the pipe **110** and may have a diameter corresponding to the pipe **110**, and the heat dissipation plate **140** may have a larger diameter than the substrate **131**. In addition, the heat dissipation plate **140** may have a larger diameter than the body **110***c* of the pipe **110**. Since the heat dissipation plate **140** may disposed at the second end **110***b* of the pipe **110**, the heat dissipation plate **140** can be formed to protrude from the second end **110***b* of the pipe **110** and can be easily formed to have a larger diameter.

(106) Further, the heat dissipation plate **140** may include a protrusion **141** protruding from a surface thereof to allow easy dissipation of heat as much as possible. The protrusion **141** may be provided in plural, whereby the surface area of the heat dissipation plate **140** can be remarkably increased thereby, as compared with the heat dissipation plate **140** not including the protrusions **141**. As the surface area of the heat dissipation plate **140** is increased, a contact area between the heat dissipation plate **140** and external air is increased, thereby enabling efficient dissipation of heat from the heat dissipation plate **140**. In one embodiment, the protrusions **141** have a plate shape extending in one direction and may be spaced apart from each other. However, it should be understood that the protrusions **141** are not limited thereto and may have any shape without limitation so long as the protrusions can increase the surface area of the heat dissipation plate **140**.
(107) In this embodiment, the heat dissipation plate **140** including the protrusions **141** may be provided as an integrated structure. With this structure, the heat dissipation plate **140** may allow easier heat transfer than a structure where the heat dissipation plate **140** is not integrally formed therewith.
(108) In the embodiment, the heat dissipation plate **140** enables easier dissipation of heat from a light source. As a result, the light source is prevented from being deteriorated due to heat from the light source, thereby providing a stable sterilization effect of the fluid treatment module **100** while improving reliability thereof.
(109) In this embodiment, since the body **110***c* is blocked at the first end **110***a* thereof and the fluid enters the pipe through one side of the pipe and is discharged from the pipe through the other side thereof, an eddy can be easily generated at a blocked portion inside the body **110***c*. A time that the fluid stays in the body **100***c* is increased by the eddy. As a result, the exposure time of the fluid to light emitted from the light source module **130** is sufficiently increased and the fluid treatment effect is also finally improved.
(110) As the time that the fluid stays in the body **100***c* is increased by the eddy, treatment efficiency with light is also improved. That is, since the fluid is irradiated with a sufficient quantity of light, a sufficient fluid treatment effect can be obtained even when the light source module **130** is disposed at one side in this embodiment. In addition, since the fluid is irradiated with a sufficient quantity of light, the body **110***c* may have a shorter length than the body according to the above embodiment. When the body **110***c* has a short distance, an eddy can be more easily generated inside the body **110***c*.
(111) Furthermore, with improvement in treatment efficiency with light, the fluid treatment module **100** according to this embodiment can omit a reflector reflecting light to travel inside the pipe **110**, for example, the reflector surrounding the inner wall of the pipe **110**, unlike the above embodiment. In this embodiment, the reflector **120** does not include the first reflector **121** (see FIG. **2** and FIG. **3**) inside the pipe **110** and may include the second reflector **123** disposed near the light source module **130**. Here, the second reflector **123** is disposed on the substrate **131** along the periphery of the mounting region of the light emitting diodes **133**. The second reflector **123**, the light emitting diode **133** has an opening exposing the mounting region of the light emitting diodes **133** and is disposed on the substrate **131** in a ring shape penetrated through upper and lower portions thereof.
(112) The fluid treatment module **100** according to this embodiment may be provided with a sealing member to tightly fasten the pipe **10** to the heat dissipation plate **140** while preventing the fluid from leaking out of the fluid treatment module.
(113) Although the fluid treatment module **100** according to the embodiment shown in FIG. **2** and FIG. **3** is provided with two sealing members, the fluid treatment module **100** according to this embodiment may be provided with a single sealing member. In this embodiment, the sealing member **150** is disposed near the fluid treatment space while surrounding an end of the transmissive window **137** in an angled-C shape.
(114) In the embodiment, the heat dissipation plate **140** may be used as a cap encapsulating the second end **110***b* of the pipe **110**. Accordingly, in this embodiment, the heat dissipation plate **140**

may have a fastening portion coupled to the pipe **110**.
(115) The fastening portion may be provided in various forms. For example, as the fastening portion, the heat dissipation plate **140** may have an insertion portion, which has a diameter corresponding to the inner diameter of the pipe **110** and is inserted into an end of the pipe **110** to encapsulate the pipe **110**.
(116) In the embodiment, the heat dissipation plate **140** is disposed at the second end **110***b* of the pipe **110** to be fastened to the pipe **110**. The heat dissipation plate **14** is formed with stepped portions having different outer diameters to be inserted into and fastened to the pipe **110**. For example, a portion of the heat dissipation plate **140** facing the second end **110***b* of the pipe **110** has an outer diameter corresponding to the inner diameter of the pipe **110**.
(117) In the embodiment, the pipe **110** and the heat dissipation plate **140** may be provided with fastening members for coupling the pipe **110** to the heat dissipation plate **140**.
(118) In one embodiment, the heat dissipation plate **140** is used not only as a heat dissipation member for dissipation of heat generated from the light source module **130**, but also as a tool for encapsulating the pipe **110**, that is, as a cap. Alternatively, an encapsulating member for encapsulating the pipe **110** may be further provided as a separate component from the heat dissipation plate **140**. In this case, the separate encapsulating member may also be formed of a material having high thermal conductivity to allow efficient dissipation of heat from the heat dissipation plate **140**.
(119) In the fluid treatment module with the above structure, the body has a short length, thereby enabling reduction in overall size thereof. In addition, the fluid treatment module secures sufficient sterilization through generation of an eddy to increase the time that the fluid stays inside the body, thereby enabling reduction in the number of light source modules and the number of light emitting diodes. As a result, the number of light source modules and the number of light emitting diodes in each light source module are reduced, thereby reducing manufacturing costs and the number of manufacturing processes, such as a process of waterproofing the opposite ends of the body or a process of assembling the heat dissipation plate.
(120) FIG. **8**A and FIG. **8**B are graphs depicting a fluid treatment effect depending upon the presence of the reflector, and FIG. **8**C is a side sectional view of a fluid treatment module used in experiments of FIG. **8**A and FIG. **8**B. In FIG. **8**A, Comparative Example is a fluid treatment module including no reflector, that is, a fluid treatment module not including the first and second reflectors, Example 1 is a fluid treatment module including the second reflector, and Example 2 is a fluid treatment module including both the first and second reflectors. In FIG. **8**B, Example 3 is a fluid treatment module including the first reflector and Example 4 is a fluid treatment module including the first and second reflectors.
(121) In FIG. **8**A and FIG. **8**B, in order to monitor the fluid treatment effect depending upon the presence of the reflector, experiments were carried out under the same conditions other than the reflector while changing the flow rate of fluid. The fluid was water and *E. coli* was used as bacteria. A bacteria inactivation degree according to treatment of the fluid was represented by a log scale (log CFU/mL) and a higher log value indicates higher sterilization capability (for example, 3 log CFU/mL indicates a bacteria inactivation degree of 99.9%). In each of the fluid treatment modules, one inlet and one outlet were disposed at the same end side to face each other, and the diameter of the inlet was two times the diameter of the outlet.
(122) Referring to FIG. **8**A, as compared with Comparative Example, Example 1 and Example 2 exhibited much higher fluid treatment efficiency, that is, much higher sterilization efficiency. For example, in treatment of 1 LPM (liter per minute) of the fluid, Comparative Example had a sterilization efficiency of 1.551 (log CFU/mL), whereas Examples 1 and 2 had a sterilization efficiency of 2.767 (log CFU/mL) and a sterilization efficiency of 3.216 (log CFU/mL), respectively. Further, in treatment of 2 LPM, Comparative Example had a sterilization efficiency of 1.080 (log CFU/mL), whereas Examples 1 and 2 had a sterilization efficiency of 2.078 (log

CFU/mL) and a sterilization efficiency of 3.732 (log CFU/mL), respectively. In particular, it could be seen that Example 2 including the first and second reflectors exhibited much higher sterilization efficiency than Example 1 including the second reflector.

(123) Referring to FIG. **8**B, Example 3 and Example 4 exhibited much higher fluid treatment efficiency, that is, much higher sterilization efficiency. For example, in treatment of 1 LPM (liter per minute) of the fluid, Examples 3 and 4 had a sterilization efficiency of 4.666 (log CFU/mL) and a sterilization efficiency of 5.413 (log CFU/mL), respectively. In particular, it could be seen that Example 4 including the first and second reflectors exhibited much higher sterilization efficiency than Example 3 including only the first reflector.

(124) As such, the fluid treatment module according to the embodiment is provided with the reflector, thereby securing much better fluid treatment capability.

(125) In the fluid treatment module according to the embodiment, the inlet and the outlet may be provided in various shapes in order to control the fluid flowing in the pipe. FIG. **9**A to FIG. **9**F are side views of fluid treatment modules each provided with an inlet and an outlet through various modifications, and FIG. **10**A to FIG. **10**F are graphs depicting fluid treatment effects (sterilization effects) of the fluid treatment modules shown in FIG. **9**A to FIG. **9**F, respectively.

(126) In FIG. **9**A to FIG. **9**F and FIG. **10**A to FIG. **10**F, experiments were carried out under the same conditions other than the locations and diameters of the inlet and the outlet, in which the flow rate of the fluid was set to 1 LPM. In particular, the fluid treatment modules used in these experiments include the reflector. The fluid was water and *E. coli* was used as bacteria. A bacteria inactivation degree according to treatment of the fluid was represented by a log scale (log CFU/mL) and a higher log value indicates higher sterilization capability (for example, 3 log CFU/mL indicates a bacteria inactivation degree of 99.9%).

(127) First, sterilization efficiency depending upon the diameter of each of the inlet and the outlet can be confirmed by comparing FIG. **10**A with FIG. **10**B.

(128) FIG. **10**A and FIG. **10**B show graphs depicting fluid treatment effects of the fluid treatment modules each including one inlet and one outlet, in which FIG. **10**A is a graph depicting the fluid treatment effect when the inlet and the outlet had the same diameter and FIG. **10**B is a graph depicting the fluid treatment effect when the diameter of the outlet was ½ the diameter of the inlet.

(129) Referring to FIG. **10**A and FIG. **10**B, the fluid treatment module including one inlet and one outlet having the same diameter exhibited a bacteria inactivation degree of 2.623 (log CFU/mL) at 1 LPM and the fluid treatment module including the inlet and the outlet having different diameters exhibited a bacteria inactivation degree of 2.350 (log CFU/mL) at 1 LPM. As such, when each of the inlet and the outlet was provided singularly, the fluid treatment module exhibited a bacteria inactivation degree of less than 3.000 (log CFU/mL), regardless of the diameter of each of the inlet and the outlet, and exhibited no significant difference in bacteria inactivation degree depending upon the diameter thereof.

(130) FIG. **10**C is a graph depicting the fluid treatment effect of a fluid treatment module including two inlets (that is, a first inlet and a second inlet) and one outlet having the same diameter as the inlets.

(131) Referring to FIG. **10**C, the fluid treatment module including two inlets and one outlet each having the same diameter exhibited a bacteria inactivation degree of 4.147 (log CFU/mL) at 1 LPM. That is, it could be seen that increase in the number of inlets provided remarkable improvement in fluid treatment effect, that is, sterilization effect, as compared with the fluid treatment module including one inlet and one outlet.

(132) It seemed that increase in the number of inlets through which the fluid flows into the pipe resulted in increase in frequency of exposure of the fluid to sterilizing light while maintaining the same flow rate by generating an eddy in the pipe, as compared with the comparative example in which the inlet was not further added. When the flow rate decreases, the frequency of exposure of the fluid to sterilizing light generally increases. However, according to the present invention, the

sterilization effect can be improved while maintaining the flow rate at 1 LPM by controlling the number of inlets, as described above.
(133) In one embodiment, sterilization efficiency of fluid treatment modules, in which the number of inlets is the same as the number of outlets and the inlet has a different diameter than the outlet, can be confirmed. FIG. **10**D is a graph depicting the fluid treatment effect of a fluid treatment module in which the number of inlets and the number of outlets were the same as those in FIG. **10**C and the diameter of the outlet was set to ½ the diameter of the inlet, and FIG. **10**E is a graph depicting the fluid treatment effect of a fluid treatment module in which the number of inlets and the number of outlets were the same as those in FIG. **10**C and the diameter of the inlet was set to ½ the diameter of the outlet. As a result, in FIG. **10**D in which the diameter of the outlet was set to ½ the diameter of the inlet, the fluid treatment module exhibited a bacteria inactivation degree of 3.888 (log CFU/mL) at 1 LPM, and, in FIG. **10**E in which the diameter of the inlet was set to ½ the diameter of the outlet, the fluid treatment module exhibited a bacteria inactivation degree of 3.897 (log CFU/mL) at 1 LPM.
(134) Referring to FIG. **10**C to FIG. **10**E, it could be seen again that, regardless of difference in diameter between the inlet and the outlet, all of the fluid treatment modules exhibited a high sterilization power of 3 (log CFU/mL) (that is, a bacteria inactivation degree of 99.9%) and the fluid treatment efficiency remarkably increases with increasing number of inlets. In addition, it could be seen that the fluid treatment efficiency could be further improved through adjustment of the diameters of the inlet and the outlet in various ways, in particular, when the inlet and the outlet were formed to have the same diameter, as shown in FIG. **10**C.
(135) FIG. **10**F is a graph depicting the fluid treatment effect of a fluid treatment module with different arrangement of the inlet and the outlet, in which the inlet and the outlet are disposed at the same end side, unlike the above experiment. Referring to FIG. **10**F, the fluid treatment module, in which the inlet and the outlet are disposed at the same end side, exhibited a bacteria inactivation degree of 1.268 (log CFU/mL) at 1 LPM, thereby indicating significant deterioration in sterilization efficiency. It seemed that the arrangement of the inlet and the outlet at the same side increased the amount of the fluid discharged from the fluid treatment module instead of sufficiently staying in the pipe.
(136) FIG. **11**, FIG. **12**A and FIG. **12**B are graphs depicting sterilization efficiency depending upon the flow rate of the fluid flowing in the pipe of the fluid treatment module according to the embodiment of the present invention. FIG. **11** is a graph depicting the fluid treatment effect (sterilization effect) of the fluid treatment module shown in FIG. **9**D, and FIG. **12**A and FIG. **12**B are graphs depicting the fluid treatment effect measured twice in use of the fluid treatment module shown in FIG. **9**C.
(137) Referring to FIG. **11**, FIG. **12**A, and FIG. **12**B, the fluid treatment module according to the embodiment exhibited very gentle decrease in overall sterilization power, despite increase in flow rate of the fluid flowing in the pipe. In other words, as shown in FIG. **11**, FIG. **12**A and FIG. **12**B, although the sterilization efficiency was gradually decreased as the fluid treatment speed was increased from 1 LPM to 5 LPM, the fluid treatment module exhibited a sterilization efficiency of 3 (log CFU/mL) or more due to insignificant reduction in sterilization power, despite increase in flow rate to five times of the initial flow rate. Accordingly, it can be seen that the fluid treatment module according to the embodiment can maintain high sterilization power despite increase in the fluid treatment speed, thereby increasing the volume of the fluid that can be sterilized thereby. In addition, the fluid treatment module according to the embodiment may be applied to a broad range of apparatuses by sufficiently ensuring sterilization power at various speeds. The fluid treatment module may be applied to various apparatuses, such as a water purifier, a washing machine, and a bidet, which may require different flow rates. As such, the fluid treatment module according to the embodiment may be applied to apparatuses requiring different flow rates.
(138) FIG. **13** is a schematic view of an apparatus, for example, a water treatment apparatus,

adopting the fluid treatment module according to the fluid treatment module according to the embodiment of the present invention.
(139) Referring to FIG. **13**, the water treatment apparatus according to the embodiment includes filters **61** primarily filtering water, a reservoir **67** storing water having passed through the filters **61**, and the fluid treatment module **100** connected to the reservoir **67**.
(140) The filters **61** serve to remove foreign matter from the supplied water. The water treatment apparatus may further include a pump (not shown) connected to the filters **61** to supply water to the filters **61**. The filters **61** may be provided in various numbers, including filters for removing large impurities, filters for removing heavy metals and bacteria, and the like. In the case of sterilizing sufficiently purified water using the sterilizing device **100**, the filters **61** may be omitted.
(141) Water from which foreign matter is removed by the filters **61** is delivered to the reservoir **67** through a connection tube **65**. The water treatment apparatus may be provided with at least one reservoir **67** or multiple reservoirs **67**. Here, in a structure where water to be purified is supplied to the water treatment apparatus, the reservoir **67** may be omitted.
(142) The fluid treatment module **100** treats water supplied from the reservoir **67**. Here, treatment in the fluid treatment module **100** may refer to various treatments, such as sterilization, purification, deodorization, and the like, as described above. As shown in the drawings, the fluid treatment module **100** may be further provided with a draw valve to allow a user to dispense water immediately.
(143) As such, with the fluid treatment module according to the present invention, it is possible to implement an apparatus having a very simple structure and high efficiency in treatment of air or water.
(144) Although some embodiments have been described herein, it should be understood that these embodiments are provided for illustration only and are not to be construed in any way as limiting the present invention, and that various modifications, changes, alterations, and equivalents can be made by those skilled in the art without departing from the spirit and scope of the invention. Therefore, the scope of the present invention is not limited to the detailed description herein and should be defined only by the accompanying claims and equivalents thereto. For example, the above embodiments may be combined in various ways without departing from the sprit and scope of the present invention.
(145) In addition, although advantageous effects provided by a certain configuration are not clearly described in description of the embodiments, it should be noted that expectable effects of the corresponding configuration should be acknowledged.

Claims

1. A fluid treatment module comprising: a pipe including a flow channel in which fluid flows, the pipe having an inlet and an outlet; a light source module comprising a substrate and at least one light emitting diode disposed in a mounting region located on an upper surface of the substrate and configured to emit light into the pipe to treat the fluid; a reflector disposed inside the pipe to reflect the light emitted from the light source module and having higher reflectivity with respect to the light than the pipe; and a heat dissipation plate arranged to be adjacent to the light source module and structured to dissipate heat from the light source module, the heat dissipation plate having higher thermal conductivity than thermal conductivity of the substrate of the light source module; wherein the reflector comprises a first reflector disposed on an inner wall of the pipe and a second reflector disposed, on the substrate and having a ring shape with an opening penetrating through a center of the ring shape and exposing the mounting region of the substrate, the opening of the second reflector has a width gradually increasing in a direction further away from the surface of the substrate.
2. The fluid treatment module according to claim 1, wherein the heat dissipation plate has a larger

area than the substrate.
3. The fluid treatment module according to claim 2, wherein the heat dissipation plate includes a metal.
4. The fluid treatment module according to claim 3, wherein the pipe further comprises a body extending in a longitudinal direction thereof, and first and second ends disposed in the longitudinal direction of the body.
5. The fluid treatment module according to claim 4, wherein the heat dissipation plate has a larger diameter than the body.
6. The fluid treatment module according to claim 1, wherein the inlet, the outlet, or both are provided in plural such that a flow speed and a flow direction of the fluid flowing into the pipe are controlled.
7. The fluid treatment module according to claim 1, wherein the reflector includes a porous material.
8. The fluid treatment module according to claim 1, wherein the reflector has a reflectivity of 80% or more.
9. The fluid treatment module according to claim 1, wherein the inlet and the outlet are provided in different numbers.
10. The fluid treatment module according to claim 9, wherein the inlet is provided as a pair and the outlet is provided singularly.
11. The fluid treatment module according to claim 10, wherein the inlet and the outlet have a same diameter.
12. The fluid treatment module according to claim 11, wherein the inlet and the outlet have different diameters.